ChemCritic 文献语料库 · 目录

生成日期 2026-10-31 | 来源：NCBI PMC 开放获取子集（E-utilities）

文献篇数	114
数值数据表	194
表格数据行	1981
反应类型	18 类
证据状态	imported_unchecked（未人工核验，不参与判定）

一、按反应类型分布

反应类型	篇数	表	行
Suzuki – Miyaura 偶联	7	10	88
Mizoroki – Heck 反应	7	20	251
Negishi 偶联	7	13	112
Miyaura 硼化 / C – H 硼化	7	10	128
C – H 活化 / 芳基化	7	10	87
转移氢化	7	11	143
CuAAC 点击化学	7	9	89
氧化 / 环氧化	7	12	93
环加成	7	11	101
烯化（Wittig 等）	7	14	122
Buchwald – Hartwig 胺化	6	11	91
Sonogashira 偶联	6	13	137
光氧化还原催化	6	8	76
烯烃复分解	6	12	133
催化氢化	6	6	81
不对称氢化	5	6	52
羰基化	5	8	84
氢胺化 / C – N 偶联	4	10	113

二、文献清单

ID	年	期刊	DOI	表	行	标题
PMC5882074	2018	Microbiology	10.1099/mic.0.000568	1	11	Recombinant expression and characterisation of the oxygen-sensitive 2-enoate r
PMC8456900	2021	Angewandte Chemie (Interna	10.1002/anie.202107267	1	9	Site‐ and Enantioselective Iridium‐Catalyzed Desymmetric Mono‐Hydrogenation of
PMC8596914	2021	Angewandte Chemie (Interna	10.1002/anie.202108503	1	9	Ru‐NHC‐Catalyzed Asymmetric Hydrogenation of 2‐Quinolones to Chiral 3,4‐Dihydr
PMC11668043	2024	Organic Letters	10.1021/acs.orglett.4c04076	1	11	Highly Enantioselective Synthesis of 3,3-Diarylpropyl Amines and 4-Aryl Tetrah
PMC12933482	2025	Precision Chemistry	10.1021/prechem.5c00087	2	12	Iridium-Catalyzed Regio- and Enantioselective Hydrogenation of 5 - Alkylidene Cy
PMC9299238	2021	Chemistry (Weinheim an Der	10.1002/chem.202102961	2	24	Rhodium Indenyl NHC and Fluorenyl‐Tethered NHC Half‐Sandwich Complexes: Synthe
PMC10366130	2023	Communications Chemistry	10.1038/s42004-023-00960-z	1	20	Transition metal-free visible light photoredox-catalyzed remote C(sp3) − H boryl
PMC10563123	2023	The Journal of Organic Che	10.1021/acs.joc.3c01534	2	17	Atroposelective Ir-Catalyzed C – H Borylation of Phthalazine Heterobiaryls
PMC12147128	2025	Journal of the American Ch	10.1021/jacs.4c17943	1	24	An Adaptive Palladium Single-Atom Catalyst Enabling Reactivity Switching betwe
PMC12293755	2025	Biomolecules	10.3390/biom15071014	2	19	Synthesis and Cytotoxicity Evaluation of Denitroaristolochic Acids: Structural
PMC12007006	2025	Journal of the American Ch	10.1021/jacs.4c18569	1	12	Trimethylaluminum Activates Zeolite-Confined Lanthanum Borohydrides to Enhance
PMC13320156	2026	Communications Chemistry	10.1038/s42004-026-02044-0	1	12	Polymeric iridium catalysts for C – H borylation of arenes under batch and conti
PMC8949195	2022	Molecules	10.3390/molecules27061999	1	10	General Method of Synthesis of 5-(Het)arylamino-1,2,3-triazoles via Buchwald – H
PMC9301730	2022	ACS Omega	10.1021/acsomega.2c01092	1	11	Phenylboronic Ester-Activated Aryl Iodide-Selective Buchwald – Hartwig-Type Amin
PMC11684481	2024	ACS Omega	10.1021/acsomega.4c07563	1	8	Microwave-Assisted Buchwald – Hartwig Double Amination: A Rapid and Promising Ap
PMC11085325	2024	Molecules	10.3390/molecules29092151	1	5	Synthesis of 5-(Aryl)amino-1,2,3-triazole-containing 2,1,3-Benzothiadiazoles v
PMC12305527	2025	RSC Advances	10.1039/d5ra02824h	3	19	Application of palladium nanoparticles supported on ordered mesoporous oxides
PMC12333095	2025	Chemical Reviews	10.1021/acs.chemrev.5c00088	4	38	Metal – N-Heterocyclic Carbene Complexes in Buchwald – Hartwig Amination Reactions
PMC9059666	2019	RSC Advances	10.1039/c8ra10212k	3	41	A new bifunctional heterogeneous nanocatalyst for one-pot reduction-Schiff bas
PMC7155033	2020	Journal of Labelled Compou	10.1002/jlcr.3805	1	5	“In‐loop”carbonylation—A simplified method for carbon‐11 labelling of drugs a
PMC9964802	2023	Molecules	10.3390/molecules28041795	2	19	Original Fluorinated Non-Isocyanate Polyhydroxyurethanes
PMC13564410	2026	Journal of the American Ch	10.1021/jacs.6c10525	1	8	Biphasic Solar Reforming of Lignin to Aromatics and CO for In Situ Carbonylati
PMC13200246	2026	Organic Letters	10.1021/acs.orglett.6c01481	1	11	Carbonylative Synthesis of β - Trifluoromethylated Heterocyclic Ketones through
PMC8928494	2022	ACS Omega	10.1021/acsomega.1c06664	3	22	Computational, Mechanistic, and Experimental Insights into Regioselective Cata
PMC11731305	2024	The Journal of Organic Che	10.1021/acs.joc.4c02266	1	11	TiO2-Catalyzed Direct Diazenylation of Active Methylene Compounds with Diazoni
PMC11528444	2024	Journal of the American Ch	10.1021/jacs.4c09621	1	12	Decarboxylative Cross-Coupling Enabled by Fe and Ni Metallaphotoredox Catalysi
PMC11811124	2025	Communications Chemistry	10.1038/s42004-025-01434-0	1	8	An active representation learning method for reaction yield prediction with sm
PMC12261849	2025	Journal of Cheminformatics	10.1186/s13321-025-01043-y	1	6	Application of the digital annealer unit in optimizing chemical reaction condi
PMC13074239	2026	Molecules	10.3390/molecules31071176	2	21	Nickel-Catalyzed Intermolecular Cyclization of 2-Bromobenzamide: A General Str
PMC13569938	2026	Chemistry (Weinheim an Der	10.1002/chem.71118	1	7	Synthesis and Functional Properties of Fluorinated Phosphonate‐Substituted Por
PMC6272186	2015	Molecules	10.3390/molecules20022837	3	20	Solvent-Free Copper-Catalyzed Azide-Alkyne Cycloaddition under Mechanochemical
PMC6274337	2016	Molecules	10.3390/molecules21121697	1	4	Structural Determinants of Alkyne Reactivity in Copper-Catalyzed Azide-Alkyne
PMC5365239	2017	Chemical Science	10.1039/c6sc02297a	1	9	Copper-catalyzed click reaction on/in live cells† †Electronic supplementary in
PMC8123656	2021	Polymers	10.3390/polym13091403	1	9	A Facile Strategy for the High Yielding, Quantitative Conversion of Polyglycol
PMC12284613	2025	Chemistry (Weinheim an Der	10.1002/chem.202501732	1	17	Double‐Click Strategy Combining CuAAC and (Thia‐) Diels‐Alder Reactions; Appli
PMC13296720	2026	The Journal of Organic Che	10.1021/acs.joc.5c02481	1	24	Mechanochemical Copper-Catalyzed Azide – Alkyne Cycloaddition (CuAAC)
PMC13435305	2026	Organic Letters	10.1021/acs.orglett.6c02540	1	6	Active Templating by Mechanochemistry

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PMC7891567	2020	Angewandte Chemie (Interna	10.1002/anie.202009704	1	7	Visible - Light - Mediated Heterocycle Functionalization via Geometrically Interru
PMC8419836	2021	The Journal of Organic Che	10.1021/acs.joc.1c00778	4	25	Preparation of Acidic 5-Hydroxy-1,2,3-triazoles via the Cycloaddition of Aryl
PMC10386021	2023	Molecules	10.3390/molecules28145372	2	26	Diastereoselective Formal 1,3-Dipolar Cycloaddition of Trifluoroethyl Amine-De
PMC11053432	2024	Molecules	10.3390/molecules29081902	1	8	Room Temperature Diels – Alder Reactions of 4-Vinylimidazoles
PMC11382162	2024	The Journal of Organic Che	10.1021/acs.joc.4c00843	1	16	Cycloadditions of 4-Alkenyl-2-aminothiazoles with Nitroalkenes in the Formal S
PMC13077698	2026	The Journal of Organic Che	10.1021/acs.joc.5c03172	1	14	Cascade Iodine-Catalyzed Synthesis of Nitrogenated Aromatics from 2 - Pyrones
PMC13185131	2026	The Journal of Organic Che	10.1021/acs.joc.6c00313	1	5	Azo-Povarov Cycloaddition of N - Carbonyl Aryldiazenes with cis,trans-Cycloocta-
PMC9086392	2018	RSC Advances	10.1039/c8ra06623j	3	34	Preparation of a novel palladium catalytic hydrogel based on graphene oxide/ch
PMC7511807	2020	Heliyon	10.1016/j.heliyon.2020.e04946	3	40	Synthesis of nano magnetic supported NHC-palladium and investigation of its ap
PMC8402173	2021	Polymers	10.3390/polym13162717	6	66	Palladium-Catalyzed Mizoroki – Heck and Copper-Free Sonogashira Coupling Reactio
PMC9101052	2022	Frontiers in Chemistry	10.3389/fchem.2022.872366	4	55	Pyrene Functionalized Highly Reduced Graphene Oxide-palladium Nanocomposite: A
PMC9476516	2022	ACS Omega	10.1021/acsomega.2c04111	1	15	Diastereoselective Synthesis of N-Methylspiroindolines by Intramolecular Mizor
PMC11557397	2024	Frontiers in Chemistry	10.3389/fchem.2024.1496234	1	12	Hybrid Pd0.1Cu0.9Co2O4 nano-flakes: a novel, efficient and reusable catalyst f
PMC13325132	2026	ACS Omega	10.1021/acsomega.6c01810	2	29	Sustainable Mizoroki – Heck Cross-Coupling Using a Pd(II)-Polymer as Precatalyst
PMC8529890	2021	ACS Omega	10.1021/acsomega.1c04826	3	32	Design of Experimental Conditions with Machine Learning for Collaborative Orga
PMC10731675	2023	Frontiers in Chemistry	10.3389/fchem.2023.1260726	4	27	Hydroamination of alkynes catalyzed by NHC-Gold(I) complexes: the non-monotoni
PMC11720983	2024	Molecules	10.3390/molecules30010004	1	16	Electrochemical α-C(sp3) – H/N – H Cross-Coupling of Isochromans and Azoles
PMC11191698	2024	Journal of the American Ch	10.1021/jacs.4c02319	2	38	One-Step Catalyst-Transfer Macrocyclization: Expanding the Chemical Space of A
PMC5838540	2018	Chemcatchem	10.1002/cctc.201701329	1	7	Hydrogenation of Lactic Acid to 1,2 - Propanediol over Ru - Based Catalysts
PMC7186356	2020	Frontiers in Chemistry	10.3389/fchem.2020.00221	1	26	Recent Advances in Ruthenium-Catalyzed Hydrogenation Reactions of Renewable Bi
PMC9224514	2022	International Journal of M	10.3390/ijms23126535	1	7	Development of High-Efficiency, Magnetically Separable Palladium-Decorated Man
PMC10870019	2023	Advanced Science	10.1002/advs.202305271	1	16	Hydrogenation of Alkynes and Olefins Catalyzed by Quaternary Ammonium Salts
PMC12177647	2025	ACS Omega	10.1021/acsomega.4c09211	1	13	Secondary Phosphine Oxide-Stabilized Pd Nanoparticles: Ligand-Controlled Chemo
PMC13265633	2026	Chemistry, an Asian Journa	10.1002/asia.70823	1	12	Synthesis of a C - 2 Functionalized l - Iduronic Acid Derivative as a Candidate Ph
PMC6644540	2018	ACS Omega	10.1021/acsomega.8b01642	1	24	In Situ-Generated Niobium-Catalyzed Synthesis of 3-Pyrroline Derivatives via R
PMC6648348	2019	ACS Omega	10.1021/acsomega.8b03027	4	53	2-Methyltetrahydrofuran as a Solvent of Choice for Spontaneous Metathesis/Isom
PMC7670115	2020	Beilstein Journal of Organ	10.3762/bjoc.16.226	2	20	Ring-closing metathesis of prochiral oxaenediynes to racemic 4-alkenyl-2-alkyn
PMC8289337	2021	Organometallics	10.1021/acs.organomet.1c00253	1	8	Routes to High-Performing Ruthenium – Iodide Catalysts for Olefin Metathesis: Li
PMC9820898	2022	International Journal of M	10.3390/ijms24010671	3	18	Ring-Opening Metathesis Polymerization and Related Olefin Metathesis Reactions
PMC9698981	2022	Molecules	10.3390/molecules27227696	1	10	Simultaneous Enantiodivergent Synthesis of Diverse Lactones and Lactams via Se
PMC5225227	2016	Monatshefte Fur Chemie	10.1007/s00706-016-1878-4	2	8	A triazine-based Ni(II) PNP pincer complex as catalyst for Kumada – Corriu and N
PMC5788718	2017	The Journal of antibiotics	10.1038/ja.2017.96	2	14	Progress Toward the Total Synthesis of Hamigerans C and D: A Direct Approach t
PMC6851999	2019	Chemistry (Weinheim an Der	10.1002/chem.201902255	1	14	Negishi's Reagent Versus Rosenthal's Reagent in the Formation of Zirconacyclop
PMC9619403	2022	Journal of the American Ch	10.1021/jacs.2c07517	1	5	Synthesis of Polycyclic Hetero-Fused 7-Deazapurine Heterocycles and Nucleoside
PMC12354725	2025	Scientific Reports	10.1038/s41598-025-14080-2	2	42	PTML models of self assembled ligand free nanoparticle catalysts for cross cou
PMC12287326	2025	Scientific Reports	10.1038/s41598-025-10180-1	1	10	Synthesis of diarylmethanes by means of Negishi cross-coupling enabled by coba
PMC13569947	2026	Chemistry (Weinheim an Der	10.1002/chem.71177	4	19	Late - Stage Modification of Halotryptophan - Containing Peptides Via Negishi Cros

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PMC8397428	2021	Organic Letters	10.1021/acs.orglett.1c02099	2	17	Cascade aza-Wittig/6 π -Electrocyclization in the Synthesis of 1,6-Dihydropyridi
PMC9964018	2023	Molecules	10.3390/molecules28041958	1	7	Wittig and Wittig – Horner Reactions under Sonication Conditions
PMC10258914	2023	Current Organic Synthesis	10.2174/1570179420666221006113032	1	8	Simple and Efficient Synthesis of Diamino Derivatives of bis-1,2,4-oxadiazole
PMC11110061	2024	Organic Process Research &	10.1021/acs.oprd.3c00414	3	14	Flow Chemistry for Synthesis of 2-(C-Glycosyl)acetates from Pyranoses via Tand
PMC10819551	2024	Molecules	10.3390/molecules29020501	3	29	Reactivity and Stability of (Hetero)Benzylic Alkenes via the Wittig Olefinatio
PMC12522154	2025	Beilstein Journal of Organ	10.3762/bjoc.21.157	3	36	α -Ketoglutaric acid in Ugi reactions and Ugi/aza-Wittig tandem reactions
PMC12728619	2025	JACS Au	10.1021/jacsau.5c01138	1	11	Beyond Wittig Olefination: Phosphorus Ylide as a Ring-Expansion Reagent for Di
PMC6941188	2019	ACS Omega	10.1021/acsomega.9b03096	1	15	Visible Light-Emitting Diode Light-Driven Cu0.9Fe0.1@RCAC-Catalyzed Highly Sel
PMC7914554	2021	Polymers	10.3390/polym13040498	1	13	Green Procedure for Aerobic Oxidation of Benzylic Alcohols with Palladium Supp
PMC8735764	2021	ACS Sustainable Chemistry	10.1021/acssuschemeng.1c02605	3	23	Efficient Syntheses of Biobased Terephthalic Acid, p-Toluic Acid, and p-Methyl
PMC11547704	2024	Molecules	10.3390/molecules29215114	1	11	A Ru-Based Complex for Sustainable One-Pot Tandem Aerobic Oxidation-Knoevenage
PMC11955942	2025	ACS Applied Materials &	10.1021/acsami.4c21206	4	21	Photocatalyzed Aerobic Oxidation of Thiols to Disulfides Using Cu2O Polyhedra
PMC12811662	2025	Angewandte Chemie (Interna	10.1002/anie.202523426	1	6	Metal‐Organic Framework/Organocatalyst Combinations as Powerful Multicatalytic
PMC13363593	2026	Nanomaterials	10.3390/nano16130835	1	4	Blue-Light-Driven Aerobic Oxidation via ROS-Generating Binuclear Cobalt(II) Co
PMC6720061	2019	Beilstein Journal of Organ	10.3762/bjoc.15.201	1	14	Naphthalene diimides with improved solubility for visible light photoredox cat
PMC7252948	2020	Journal of the American Ch	10.1021/jacs.0c00629	2	14	Umpolung Difunctionalization of Carbonyls via Visible-Light Photoredox Catalyt
PMC8179447	2021	Chemical Science	10.1039/d0sc06306a	1	7	Photoredox catalysis on unactivated substrates with strongly reducing iridium
PMC11020166	2023	Organic Letters	10.1021/acs.orglett.3c02389	2	31	Decarboxylative Nickel- and Photoredox-Catalyzed Aminocarbonylation of (Hetero
PMC12340967	2025	Organic Letters	10.1021/acs.orglett.5c02079	1	6	Access to SCF3 - Substituted Indolizines via a Photocatalytic Late-Stage Functio
PMC12856654	2025	ACS Physical Chemistry Au	10.1021/acsphyschemau.5c00103	1	4	When Better Quenching Means Lower Yields: Electrostatic Control of Cage Escape
PMC6017456	2018	Molecules : A Journal of S	10.3390/molecules23010154	1	6	A Mild Aqueous Sonogashira Reaction as a Fluorescent Labeling Strategy for 5-B
PMC9073152	2019	RSC Advances	10.1039/c9ra07393k	1	5	High-yielding Pd2(dba)3 · C6H6-based four-fold Sonogashira coupling with selenop
PMC10254696	2023	Molecules	10.3390/molecules28114364	1	17	A General Synthesis of Cross-Conjugated Enynones through Pd Catalyzed Sonogash
PMC11667721	2024	Organic Letters	10.1021/acs.orglett.4c03519	1	19	Palladium-Catalyzed Carbonylative Sonogashira Coupling of Aliphatic Alkynes an
PMC13097261	2026	The Journal of Organic Che	10.1021/acs.joc.6c00362	4	58	N - Mesyl-Enabled Cu2O Catalysis: Synthesis of (E) - 3-Alkylideneisoindolin-1-ones
PMC13158381	2026	ChemistryOpen	10.1002/open.70217	5	32	Recent Advances in Nickel Catalysts for Sonogashira Coupling
PMC6222659	2018	Molecules : A Journal of S	10.3390/molecules23102435	3	22	Efficient Suzuki – Miyaura C-C Cross-Couplings Induced by Novel Heterodinuclear
PMC9079834	2018	RSC Advances	10.1039/c8ra01819g	1	15	Extension and functionalization of an encapsulating macrobicyclic ligand using
PMC9075783	2019	RSC Advances	10.1039/c9ra07044c	1	12	Scope and limitation of propylene carbonate as a sustainable solvent in the Su
PMC11199998	2024	Advanced Science	10.1002/advs.202309706	1	12	Palladium‐Catalyzed Atroposelective Suzuki – Miyaura Coupling to Construct Axial
PMC11434556	2024	Molecules	10.3390/molecules29184398	2	14	A Ligand-Free Approach towards Coumarin Analogs via Natural Deep Eutectic Solv
PMC13010367	2026	Organic Process Research &	10.1021/acs.oprd.5c00298	1	6	Extracting Mechanistic Information from an Open Data Set for a Pharma-Relevant
PMC13010305	2026	Organic Process Research &	10.1021/acs.oprd.5c00443	1	7	A Concise Process to Lenacapavir Sodium: Demonstration of One-Pot Sequential H
PMC6641154	2017	ACS Omega	10.1021/acsomega.7b00545	1	5	Nickel-Catalyzed Transfer Hydrogenation of Benzonitriles with 2-Propanol and 1
PMC6207922	2018	Angewandte Chemie (Interna	10.1002/anie.201808800	1	16	FLP‐Catalyzed Transfer Hydrogenation of Silyl Enol Ethers
PMC7450644	2020	ACS Omega	10.1021/acsomega.0c02957	1	22	General and Efficient Copper-Catalyzed Oxazaborolidine Complex in Transfer Hyd
PMC8697831	2021	RSC Advances	10.1039/d1ra01102b	2	27	Copper(i) pyrimidine-2-thiolate cluster-based polymers as bifunctional visible

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PMC11459429	2024	The Journal of Organic Che	10.1021/acs.joc.4c01701	1	15	Iron-Catalyzed Transfer Hydrogenation of Allylic Alcohols with Isopropanol
PMC11279712	2024	Molecules	10.3390/molecules29143420	2	27	Ru(II)-Catalyzed Asymmetric Transfer Hydrogenation of α-Alkyl-β-Ketoaldehydes
PMC12609509	2025	Molecules	10.3390/molecules30214212	3	31	Regioselective Transfer Hydrogenation of Substituted Oxiranes with Alcohols Us

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